Appendix to DNB Analysis“Healthy markets, healthy growth: Dynamics, market power and misallocation in the Dutch business sector”

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A Data and Estimation of Markups, Markdowns, and Misallocation

In this section, we describe the CBS microdata used and the estimation of markups, markdowns, and misallocation. The microdata are accessible exclusively within the secure CBS microdata environment (see: https://www.cbs.nl/microdata).

A.1 Microdata Used

All datasets used are derived from confidential microdata of Statistics Netherlands (CBS), which are made available to accredited research institutions through the secure CBS research environment. The dataset covers the period 2007–2023 and, in line with the main text of the DNB Analysis, focuses on the delineated market sector.

The Statistics on the Finances of Non-Financial Corporations (NFO) form the core of our dataset. The NFO is a CBS-compiled microdata source with detailed annual information on the balance sheet and profit-and-loss accounts of virtually all non-financial corporations with legal personality (private and public limited companies) in the Netherlands. The data are based on a combination of primary surveys of large firms (SFGO) and administrative data from corporate income tax filings of smaller firms (SFKO). The NFO allows us to consistently measure firms' financial variables, including turnover, value added, capital, and intermediate inputs.

Information on labor input is not available in the NFO and is therefore obtained from the General Business Register (ABR) of CBS. The ABR contains core information for all active firms, such as industry classification at a detailed SBI-4D level and the number of workers. Using the enterprise group identification number (`ond-id`), which is consistent across the NFO and ABR, we link financial data from the NFO to labor data from the ABR at the enterprise group level. This combination enables us to analyze markups, markdowns, and misallocation of labor and capital for a large and consistent population of firms. In line with Adema et al. (2015), we exclude firms with fewer than two workers. Per year, our sample contains data on approximately 300,000 enterprise groups.

A.2 Markups

The markup is defined as the ratio of the output price to marginal cost (De Loecker en Warzynski, 2012). Estimating markups is not only relevant as an indicator of product market power, but also as a building block for the estimation of markdowns. Once marginal costs can be approximated via the markup, one can also quantify the wedge between the marginal revenue product of labor and actual wage costs. As such, the estimation of markups directly links to our estimation of markdowns (see Section A.3) and misallocation (see Section A.4).

To compute the markup following the method of De Loecker en Warzynski (2012), we assume the following production function:

$$Y_{is,t} = Y_{is,t}(M_{is,t}, L_{is,t}, K_{is,t}; \omega_{is,t}) \,, \tag{1}$$

where $Y_{is,t}$ denotes output, $L_{is,t}$ labor, $K_{is,t}$ the sum of the volume of tangible and intangible fixed assets, $M_{is,t}$ the sum of the volume of raw materials, intermediate inputs, and other operating expenses, and $\omega_{is,t}$ firm-level productivity for firm i in sector s in year t.

The first-order condition for the variable/flexible input, assumed to be free of adjustment costs ($M_{is,t}$), is:

$$\mu_{is,t} = \frac{p^Y_{is,t}}{\lambda_{is,t}} = \theta^M_{is,t} * \frac{p^Y_{is,t} Y_{is,t}}{p^m_{is,t} M_{is,t}}. \tag{2}$$

Here, $\mu_{is,t}$ denotes the markup of firm i in sector s in year t, $\lambda_{is,t}$ marginal cost, $\theta^M_{is,t}$ the output elasticity of the flexible input, and $\frac{p^Y_{is,t} Y_{is,t}}{p^m_{is,t} M_{is,t}}$ the ratio of revenues to expenditures on that input.

Both $p^Y_{is,t} Y_{is,t}$ and $p^m_{is,t} M_{is,t}$ are directly observed by CBS. Empirically estimating the markup therefore hinges on estimating the output elasticity $\theta^M_{is,t}$. We do so in two ways. The first method—which we use in the main text of the DNB Analysis—is based on the Olley–Pakes production function approach and is described in Section A.2.1. In Section A.2.2, we describe the alternative approach we estimate for the markup.

A.2.1 Estimating the Output Elasticity Following Olley en Pakes (1996)

In the DNB Analysis, markups are computed using a sector-specific Cobb–Douglas production function:

$$Y_{is,t} = \omega_0^{\beta_0} K^{\beta_k}_{is,t} L^{\beta_l}_{is,t} M^{\beta_m}_{is,t} e^{\varepsilon_{is,t}}. \tag{3}$$

Here, ω_0 denotes a sector-specific constant, and $\varepsilon_{is,t}$ is an unobservable firm-, sector-, and time-specific factor affecting a firm's production volume (such as a productivity shock).

Taking natural logarithms yields the following log-linear specification:

$$y_{is,t} = \beta_0 + \beta_k k_{is,t} + \beta_l l_{is,t} + \beta_m m_{is,t} + \varepsilon_{is,t}. \tag{4}$$

This equation is estimated separately for the 25% largest and 75% smallest firms, measured by turnover, allowing for differences in production technologies between large and small firms. To obtain real variables, we use sector-specific deflators for turnover, investment, and intermediate inputs, in line with van Heuvelen et al. (2019).

Our primary interest lies in β_m in equation (4), as this parameter directly enters the firm-level markup calculation. While we distinguish between large and small firms, we assume that this elasticity is otherwise constant within sectors and over time.

Equation (4) could in principle be estimated using ordinary least squares (OLS), but this leads to endogeneity problems. Firms' choices of capital, labor, and materials typically depend on unobserved factors that also affect productivity. If producers have (partial) information about their productivity at the time input decisions are made, the regressors will be correlated with the error term. This violates the exogeneity assumption and results in upward-biased OLS estimates (Ackerberg et al., 2015).

To make this explicit, the error term ($\epsilon_{is,t}$) is decomposed into a predictable productivity component ($\upsilon_{is,t}$) and an unanticipated shock ($\xi_{is,t}$):

$$y_{is,t} = \beta_0 + \beta_k k_{is,t} + \beta_l l_{is,t} + \beta_m m_{is,t} + \upsilon_{is,t} + \xi_{is,t}, \tag{5}$$

where $\upsilon_{is,t}$ represents productivity that is (partially) observed by the producer. Crucially, input choices depend on $\upsilon_{is,t}$, rendering OLS inconsistent.

Olley en Pakes (1996) propose a solution that exploits differences in input flexibility and information contained in investment decisions. Materials are assumed to be fully flexible within period t, whereas capital and labor result from decisions made in period $t-1$. Productivity is assumed to evolve according to a first-order Markov process, and investment is assumed to contain information about current productivity. Under these assumptions, the investment decision can be written as

$$i_{is,t} = f_t(k_{is,t}, l_{is,t}, \upsilon_{is,t}), \tag{6}$$

which can be inverted to yield

$$\upsilon_{is,t} = f_t^{-1}(k_{is,t}, i_{is,t}, l_{is,t}). \tag{7}$$

Substituting this expression into equation (5) yields:

$$y_{is,t} = \beta_0 + \beta_k k_{is,t} + \beta_l l_{is,t} + \beta_m m_{is,t} + f_t^{-1}(k_{is,t}, i_{is,t}, l_{is,t}) + \xi_{is,t}. \tag{8}$$

Because the functional form of $f_t^{-1}(\cdot)$ is unknown, it is approximated by a flexible polynomial in capital, investment, and labor. Although this induces collinearity with the quasi-fixed inputs, the output elasticity of materials β_m remains identifiable, because material inputs are not included in the polynomial. For the purpose of estimating the markup, a consistent estimate of β_m suffices; the remaining input elasticities can be identified in a subsequent step.

A drawback of the Olley–Pakes method is that it requires firm-level investment data. Observations with zero or negative investment are therefore dropped, which can reduce sample size and lead to selection bias.

A.2.2 Estimating the Output Elasticity Following De Loecker et al. (2026)

De Loecker et al. (2026) develop a method to estimate markups without explicitly estimating a firm-level production function. This is attractive because production function estimation is empirically challenging. Unobserved productivity shocks affect both output and input choices, leading to endogeneity. Moreover, in the presence of market imperfections, input prices do not necessarily reflect marginal products, meaning output elasticities cannot be interpreted as purely technological parameters. The proposed approach circumvents these issues by deriving markups from the optimal choice of flexible inputs, without requiring additional firm-level investment information.

The core of this approach is that firms choose their flexible inputs such that marginal revenue equals marginal cost, adjusted for market power. Under this assumption, the output elasticity of the flexible input can be approximated using its cost share. Specifically, the output elasticity is estimated at the sector-year level as:

$$\theta_{s,t}^{M} = \text{median}\left(1.1 * \frac{p_{is,t}^{m} M_{is,t}}{p_{is,t}^{C} C_{is,t}}\right), \tag{9}$$

where $p_{is,t}^{C} C_{is,t}$ denotes total production costs of firm i in year t, and $p_{is,t}^{m} M_{is,t}$ expenditures on the flexible input. The factor 1.1 reflects the assumption of modest economies of scale of this magnitude. By estimating the elasticity at the sector-year level, it is allowed to vary across sectors and over time.

In contrast to the method of Olley en Pakes (1996), which assumes constant output elasticities within sectors, this approach allows technological characteristics to evolve gradually over time. Given the estimated output elasticity, the firm-level markup follows directly from the ratio of revenues to expenditures on the flexible input:

$$\mu_{is,t} = \theta^M_{s,t} * \frac{p^Y_{is,t} Y_{is,t}}{p^m_{is,t} M_{is,t}}. \tag{10}$$

A.2.3 Decomposition

To shed light on the sources of changes in markups at the macro level, we employ a firm-level shift–share decomposition as proposed by De Loecker et al. (2020). This decomposition allows changes in the average markup to be broken down into contributions from within-firm adjustments, shifts in market shares across firms, and entry and exit dynamics. As such, it aligns with the broader literature that interprets macro-level developments in market power as the outcome of micro-level reallocation processes.

The change in the revenue-weighted average markup in the economy can be expressed as:

$$\begin{aligned}
\Delta\bar{\mu}_t = & \underbrace{\sum_i (\mu_{i,t} - \mu_{i,t-1}) a_{i,t-1}}_{\text{within effect}} \\
& + \underbrace{\underbrace{\sum_i (a_{i,t} - a_{i,t-1})(\mu_{i,t-1} - \bar{\mu}_{t-1})}_{\text{market share}} + \underbrace{\sum_i (a_{i,t} - a_{i,t-1})(\mu_{i,t} - \mu_{i,t-1})}_{\text{cross term}}}_{\text{between effect}} \\
& + \underbrace{\sum_{i \in \text{entry}} (\mu_{i,t} - \bar{\mu}_{t-1}) a_{i,t} - \sum_{i \in \text{exit}} (\mu_{i,t-1} - \bar{\mu}_{t-1}) a_{i,t-1}}_{\text{net entry effect}}.
\end{aligned} \tag{11}$$

Here, $\mu_{i,t}$ denotes the (unweighted) markup of firm i in period t, $\bar{\mu}_t$ the revenue-share-weighted average markup across the entire economy, and $a_{i,t}$ the revenue share of firm i at time t.

The first term, the *within effect*, captures changes in markups within incumbent firms, weighted by their initial market shares. This component reflects, for example, price adjustments or cost structure changes among incumbents, conditional on their existing market positions.

The *between effect* measures the extent to which changes in the average markup are driven by shifts in market shares across firms. This effect can be decomposed into a pure market share component—where firms with relatively high or low markups gain or lose market share—and a cross term measuring the extent to which changes in market shares coincide with changes in firm-level markups. This component aligns with the literature linking market power to the reallocation of economic activity toward firms with higher markups.

Finally, the *net entry effect* captures the contribution of entering and exiting firms. Entry reduces (increases) the average markup when entrants have lower (higher) markups than the incumbent average, while exit has the opposite effect. This term therefore reflects the role of selection and creative destruction, as emphasized in the dynamic literature on market power and productivity.

A.3 Markdowns

Where the markup measures the wedge between price and marginal cost in the output market, the markdown measures the wedge between the marginal revenue product of labor and the actual compensation of labor in the labor market. This aligns with the misallocation framework, which focuses on differences in marginal revenue products across firms.

For the estimation of markdowns, we follow Yeh et al. (2022). Their starting point is that, once the markup has been estimated using a suitable flexible input, marginal production costs can be approximated. Combined with an estimate of the output elasticity of labor, this allows the ratio between the marginal revenue product of labor and actual wage costs to be derived. Under the assumption that labor is chosen statically and does not involve adjustment costs, the markdown is given by:

$$\nu_{is,t} = \frac{\theta^L_{s,t}}{\mu_{is,t}} * \frac{p^Y_{is,t} Y_{is,t}}{w_{is,t} L_{is,t}}. \tag{12}$$

Here, $\nu_{is,t}$ denotes the markdown of firm i in sector s and year t, $\mu_{is,t}$ the previously estimated markup, $\theta^L_{s,t}$ the output elasticity of labor, and $\frac{p^Y_{is,t} Y_{is,t}}{w_{is,t} L_{is,t}}$ the ratio of revenues to wage costs.

The intuition is that under monopsonistic buying power in the labor market, the marginal wage cost increases with the amount of labor hired. The first-order condition for labor can then be written as:

$$w'_{is,t}(L_{is,t}) L_{is,t} + w_{is,t}(L_{is,t}) = \lambda_{is,t} \frac{\partial Y_{is,t}(\cdot)}{\partial L_{is,t}}, \tag{13}$$

where $\lambda_{is,t}$ represents the shadow value of production for firm i. As a result, the marginal revenue product of labor may exceed the actual wage rate. Rewriting this condition and using the definition of the markup yields equation (12).

In this analysis, the markdown $\nu_{is,t}$ is computed based on the markup estimate from Section A.2. The output elasticity of labor $\theta^L_{s,t}$ is estimated using the production function approach of Olley en Pakes (1996), while the ratio of revenues to wage costs is directly observed in the data.

In practice, however, it is plausible that labor input is not fully static. Expanding or contracting the workforce is typically associated with search costs, administrative burdens, and firing costs. As a result, the wedge between marginal revenue and wage costs reflects not only employer market power, but also labor adjustment costs. To correct for this, we follow Yeh et al. (2022) and apply a correction to the initially estimated markdowns. In the presence of adjustment costs, the following holds:

$$\frac{R'_{is,t}(L^*_{is,t})}{w_{is,t}(L^*_{is,t})} = \nu_{is,t} + A_{is,t}(L^*_{is,t}, L_{is,t-1}), \tag{14}$$

where $A_{is,t}(\cdot)$ captures the contribution of adjustment costs. Assuming quadratic adjustment costs,

$$\Phi(L_{is,t}, L_{is,t-1}) = \frac{\gamma}{2}\left(\frac{L_{is,t} - L_{is,t-1}}{L_{is,t-1}}\right)^2, \tag{15}$$

with $\gamma \geq 0$, the uncorrected markdown $\hat{\nu}_{is,t}$ can be adjusted as follows:

$$\nu_{is,t} = \frac{\hat{\nu}_{is,t} - \gamma * \left(g^L_{is,t}(1 + g^L_{is,t}) - \beta g^L_{is,t+1}(1 + g^L_{is,t+1})(1 + g^w_{is,t+1})\right)}{1 + \frac{\gamma}{2}(g^L_{is,t})^2}. \tag{16}$$

We set the discount factor β equal to 1 and the adjustment cost parameter γ equal to 0.185, in line with Yeh et al. (2022). Growth in hours worked g^L and growth in total wage costs g^w are computed at the firm level using the available data.

A.4 Misallocation

To estimate the effect of misallocation of resources within industries on aggregate total factor productivity, we use the framework of Hsieh en Klenow (2009), which has recently been applied by, among others, Gopinath et al. (2017), Gamberoni et al. (2016), Calligaris (2015), and Calligaris et al. (2018). The Hsieh–Klenow model shows that frictions that distort the marginal revenue products of capital and labor reduce aggregate factor productivity.

In the Hsieh–Klenow model, aggregate output is described by a Cobb–Douglas production technology, while at the industry level output is a CES (constant elasticity of substitution) aggregate of differentiated products. For comparability with Hsieh en Klenow

(2009) and Gopinath et al. (2017), we set the elasticity of substitution σ in the CES function equal to 3.

Each individual firm produces its unique good according to a standard Cobb–Douglas production function:

$$Y_{is,t} = \omega_{is,t} K_{is,t}^{\alpha_s} L_{is,t}^{1-\alpha_s}. \tag{17}$$

We measure nominal firm-level value added, $p_{is,t}^{Y} Y_{is,t}$, as gross revenues minus the costs of materials used in production. Since we do not observe firm-level prices, we use two-digit industry deflators, following, for example, Dias et al. (2016), Gorodnichenko et al. (2018), and Gopinath et al. (2017). In practice, this implies that we approximate $Y_{is,t}$ as $p_{is,t}^{Y} Y_{is,t}/p_{s,t}^{Y}$, where $p_{s,t}^{Y}$ is the industry deflator.[1]

We measure labor input $L_{is,t}$ as the firm's wage bill, deflated using the same two-digit industry deflators applied to nominal output, i.e. as $W_{is,t}L_{is,t}/p_{s,t}^{Y}$. We use the nominal wage bill rather than employment to correct for differences in workforce quality across firms, as argued in, among others, Hsieh en Klenow (2009) and Gopinath et al. (2017). Finally, we measure firm-specific real capital stock $K_{is,t}$ using the book value of tangible fixed assets, deflated by the aggregate deflator for gross fixed capital formation of non-financial corporations, p_t^K. This implies that we approximate $K_{is,t}$ as $p_{is,t}^{K} K_{is,t}/p_t^K$.

Each firm behaves as a monopolist in its differentiated product and is subject to two types of distortions: a capital wedge $\tau_{is,t}^{K}$, which alters the relative marginal revenue product of capital versus labor, and an output wedge $\tau_{is,t}^{Y}$, which distorts the marginal products of capital and labor proportionally. Market distortions enter the firm's profit function as:

$$\pi_{is,t} = (1 - \tau_{is,t}^{Y}) p_{is,t}^{Y} Y_{is,t} - w_{s,t} L_{is,t} - (1 + \tau_{is,t}^{K}) r_{s,t} K_{is,t}. \tag{18}$$

The first-order conditions for profit maximization with respect to labor and capital are:

$$MRPL_{is,t} = p_{is,t}^{Y} \frac{\partial Y}{\partial L} = (1-\alpha_s) \left(\frac{\sigma - 1}{\sigma}\right) \left(\frac{p_{is,t}^{Y} Y_{is,t}}{L_{is,t}}\right) = \left(\frac{1}{1 - \tau_{is,t}^{Y}}\right) w_{s,t}, \tag{19}$$

$$MRPK_{is,t} = p_{is,t}^{Y} \frac{\partial Y}{\partial K} = \alpha_s \left(\frac{\sigma - 1}{\sigma}\right) \left(\frac{p_{is,t}^{Y} Y_{is,t}}{K_{is,t}}\right) = \left(\frac{1 + \tau_{is,t}^{K}}{1 - \tau_{is,t}^{Y}}\right) r_{s,t}. \tag{20}$$

We estimate $MRPL$ and $MRPK$ using firm-level data on value added, labor, and capital. In addition, we compute α_s as one minus the share of wage costs in value added at the five-digit industry level. The elasticity of substitution between varieties is assumed to be

[1] Industry deflators are publicly available data from Statistics Netherlands.

the same for all firms, sectors, and years and is set equal to $\sigma = 3$.

Following Foster et al. (2008), the Hsieh–Klenow model distinguishes between revenue productivity (TFPR) and physical productivity (TFPQ). TFPQ can be defined by rewriting equation (17):

$$TFPQ_{is,t} = \omega_{is,t} = \kappa_{s,t} \frac{(p^Y_{is,t} Y_{is,t})^{\frac{\sigma}{\sigma-1}}}{K^{\alpha_s}_{is,t} L^{1-\alpha_s}_{is,t}}, \tag{21}$$

where $\kappa_{s,t} = (p^Y_{s,t} Y_{s,t})^{\frac{1}{\sigma-1}} / p^Y_{s,t}$ is an industry-specific scaling factor. Because firm-level prices are not directly observed, we do not directly observe real output at the firm level, only nominal output $p^Y_{is,t} Y_{is,t}$. Assuming iso-elastic inverse demand,[2] we therefore raise $p^Y_{is,t} Y_{is,t}$ to the power $\sigma/(\sigma - 1)$ to obtain a measure of physical output.

TFPR is then defined as:

$$TFPR_{is,t} = p^Y_{is,t} \omega_{is,t} = \left(\frac{\sigma}{\sigma - 1}\right) \left(\frac{MRPK_{is,t}}{\alpha_s}\right)^{\alpha_s} \left(\frac{MRPL_{is,t}}{1 - \alpha_s}\right)^{1-\alpha_s}. \tag{22}$$

In the model of Hsieh en Klenow (2009), TFPR does not vary across firms within an industry unless firms face output and/or capital distortions.[3]

Within industries, large differences in the marginal revenue product of capital can arise when capital remains stuck for prolonged periods in low-productivity firms. Under efficient allocation, these firms would shrink, releasing capital to more productive firms, until marginal revenue products of capital are equalized within the sector. A similar intuition applies to labor. Recent studies link wedges in the labor first-order condition to labor market power, see for example Dobbelaere en Kiyota (2018) and Mertens (2020). Empirically, we indeed observe both negative and positive labor wedges.

Following Hsieh en Klenow (2009), we estimate the impact of misallocation on the level of TFPQ by defining the "efficient" level of TFPQ as the level that would prevail under an allocation with no dispersion in $MRPK$, $MRPL$, and $TFPR$, such that $TFPR_{is,t} = \overline{TFPR}_{s,t}$, where:

$$\overline{TFPR}_{s,t} = \frac{p^Y_{s,t} Y_{s,t}}{K^{\alpha_s}_{s,t} L^{1-\alpha_s}_{s,t}}. \tag{23}$$

It can then be shown that the resulting difference in $\log(TFPQ)$—the misallocation gain $\Lambda_{s,t}$—can be expressed as a function of the variables introduced in equations (17)–(23).

[2] Haltiwanger et al. (2018) provide a critical assessment of this assumption and show that part of the observed dispersion in TFPR can be attributed to demand shocks.

[3] Zero dispersion in MRPK, MRPL, and TFPR only requires that the wedges be identical across firms within a sector; a distortion may still exist if it affects all firms equally.

See, for example, equations (1)–(11) in Gopinath et al. (2017) for a detailed derivation.

Note that equation (23) accounts only for misallocation within industries. However, Calligaris et al. (2018) show that this within-industry component is typically large. Applying their TFPR variance decomposition (see equation (2) in Calligaris et al. (2018)), we find a very similar result for the Netherlands: on average over the sample period, 80% of TFPR dispersion is attributable to the within component.

Our misallocation measure represents an upper bound on within-industry misallocation, since all variation in marginal revenue products is attributed to misallocation. This is a strong assumption. The assumptions underlying the Hsieh en Klenow (2009) model are quite stringent: all firms within a sector share the same markup and capital intensity, there are no adjustment costs, and the elasticity of substitution between capital and labor follows from a Cobb–Douglas production function with constant returns to scale. Moreover, input composition effects are abstracted from. For example, differences in the mix of high- and low-skilled labor with different marginal productivities can generate dispersion in $MRPL$, even when allocation is in fact efficient. To the extent that such composition effects are constant over time, changes in $MRPL$ can still be interpreted as changes in misallocation. Despite these limitations, we consider the model sufficiently suitable for mapping misallocation in the economy.

B Regression Analysis: Dependent Variable Markups

In this section, we describe the regression analysis underlying the finding that an increase in domestic market concentration does not necessarily have negative effects, and that in internationally competitive industries higher concentration is often associated with selection and scale effects without translating into higher markups (see the Summary and Chapter 2 of the DNB Analysis).

Table I presents the results of our fixed-effects regression analysis, in which we relate sector-year (SBI-3D) markups to the entry rate (entries), the exit rate (exits), and the markdown.

The baseline regression results are reported in column (1) of Table I. This specification includes both year and sector fixed effects. The (adjusted) R^2 of this regression is approximately 0.86.[4] For comparison, column (4) shows results from a regression that includes only year fixed effects. In this specification, a large share of the explanatory power disappears and the (adjusted) R^2 drops to about 0.20. This underscores the importance of including industry dummies and, more generally, controlling for unobserved heterogeneity across sectors.

Regression (1) shows that both higher market concentration (HHI) and a higher markdown, conditional on year and sector fixed effects, have a statistically significant negative effect on the markup. These results are highly robust: the estimated effects remain statistically significant across all specifications in columns (1) through (6), both in models with year and sector fixed effects and in models with sector fixed effects only.

In regressions (2) and (3), we examine whether the effect of market concentration on markups differs across industries with varying degrees of domestic orientation. To do so, we use the taxonomy developed by Calligaris et al. (2024), which determines for each industry the share of sales on the domestic, European, and global markets. Based on this taxonomy, we define two variables. The variable *Shielded industry (dummy)* equals one if the domestic sales share exceeds the median value in our sample (0.81), and zero otherwise. The variable *Shielded industry (share)* is the continuous counterpart and captures the share of sales on the domestic market.

Columns (2) and (3) report the results of regressions in which, respectively, the dummy and the continuous measure of domestic orientation are interacted with market concentration. The results show that an increase in market concentration has a negative effect on markups, but that this effect depends on the degree of international orientation of the in-

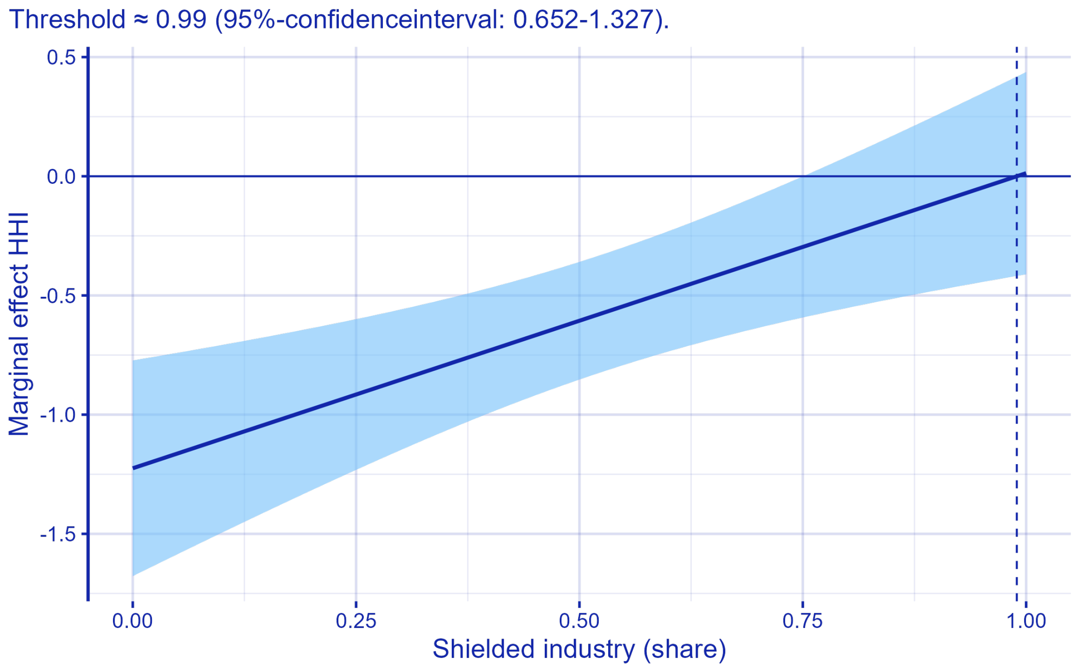
[4] Minor differences are due to rounding in the table.

Tabel I: Regression results: markup

	(1)	(2)	(3)	(4)	(5)	(6)
Market concentration (HHI)	−0.454**	−0.792***	−0.425**	−0.559**	−0.535*	−0.558**
	(0.169)	(0.185)	(0.134)	(0.199)	(0.242)	(0.195)
Births	−0.233*	−0.219*	−0.230*	0.467*	0.477*	0.504*
	(0.107)	(0.107)	(0.107)	(0.218)	(0.216)	(0.215)
Deaths	−0.177	−0.143	−0.156	0.719	0.728	0.669
	(0.208)	(0.206)	(0.204)	(0.435)	(0.442)	(0.445)
Markdown	−0.034**	−0.034**	−0.033**	−0.051***	−0.052***	−0.053***
	(0.011)	(0.011)	(0.011)	(0.012)	(0.012)	(0.012)
Shielded sector (dummy)					0.000	
					(0.028)	
Shielded sector (share)						0.037
						(0.039)
Marketconc. (HHI) × Shielded sector (dummy)		0.871**			−0.076	
		(0.281)			(0.408)	
Markteconc. (HHI) × Shielded sector (sector)			1.238**			−0.582
			(0.370)			(0.785)
Num.Obs.	1789	1789	1789	1795	1795	1795
R2	0.816	0.819	0.819	0.208	0.208	0.212
R2 Adj.	0.801	0.804	0.803	0.200	0.199	0.203
R2 Within	0.062	0.078	0.074	0.204	0.204	0.208
R2 Within Adj.	0.059	0.075	0.071	0.202	0.201	0.205
AIC	−4394.3	−4423.3	−4416.4	−2004.2	−2000.8	−2010.0
BIC	−3614.8	−3638.3	−3631.4	−1894.3	−1880.0	−1889.2
RMSE	0.07	0.06	0.07	0.14	0.14	0.14
Std.Errors	by: sector	by: sector	by: sector	by: sector	by: sector	by: sector
FE: year	X	X	X	X	X	X
FE: sector	X	X	X			

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Figuur 1: Interaction effect Market Concentration (HHI) × Shielded industry (share)



dustry. The interaction terms *Market concentration (HHI)* × *Shielded industry (dummy)* and *Market concentration (HHI)* × *Shielded industry (share)* are positive and statistically significant, indicating that the negative effect of market concentration on markups is weaker—and for very strongly domestically oriented industries even reverses sign.

Figure 1 illustrates how these results can be interpreted. The figure provides a graphical representation of regression (3) and shows the marginal effect of market concentration on markups for different levels of domestic orientation. The point estimate suggests that the effect becomes positive at a domestic orientation of approximately 0.99 (indicated by the vertical dashed line). This implies that only for industries that are almost entirely (around 99%) oriented toward the domestic market does an increase in market concentration coincide with higher markups. For industries with lower domestic orientation, higher market concentration does not lead to higher markups.

Figure 1 also shows the uncertainty surrounding these estimates. The upper bound of the 95% confidence interval crosses the zero line at a domestic orientation of around 0.75. Taking this uncertainty into account, it is possible that for industries with more than roughly 75% domestic orientation market concentration has a positive effect on markups. At the same time, the effect may remain negative across all levels of domestic orientation, since the lower bound of the confidence interval lies below zero for all values.

C Extra Figures

This chapter contains the additional figures referred to in the main text of the DNB Analysis.

Figure II: Internal en external dynamics dynamics per country

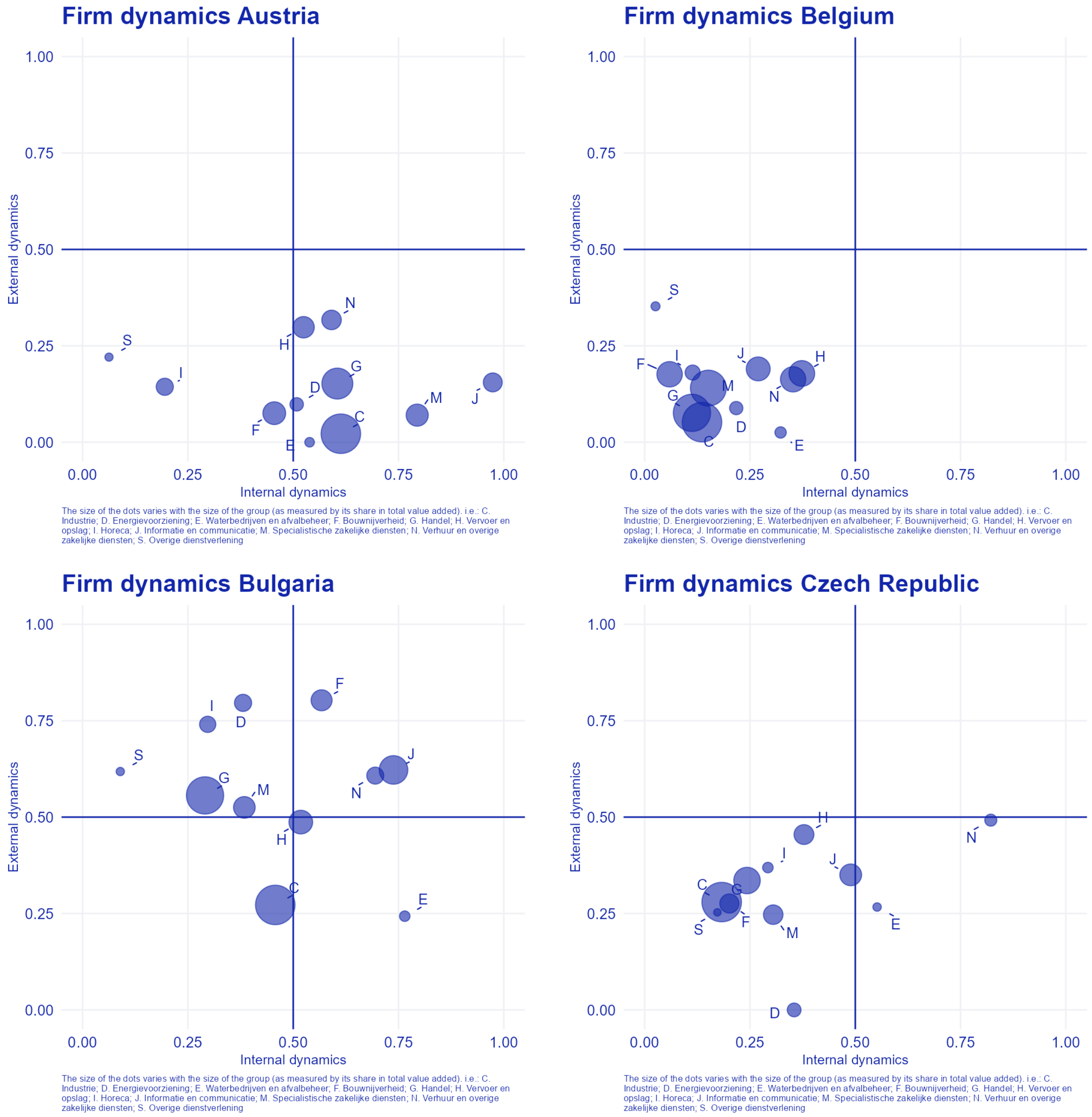


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Figure II: Internal en external dynamics dynamics per country (continued)

Figuer continued on next page

Figure II: Internal en external dynamics dynamics per country (continued)

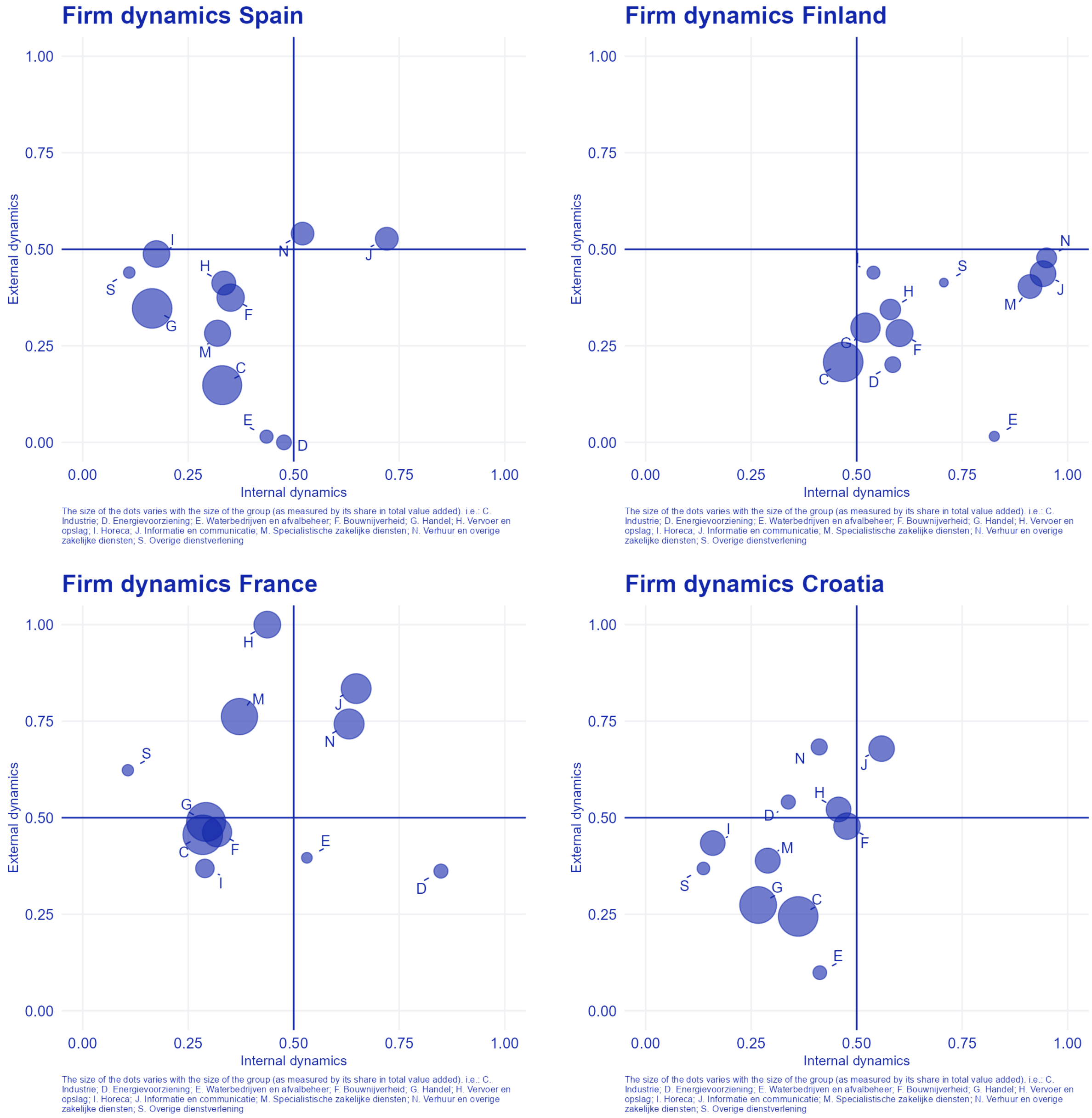

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Figure II: Internal en external dynamics dynamics per country (continued)

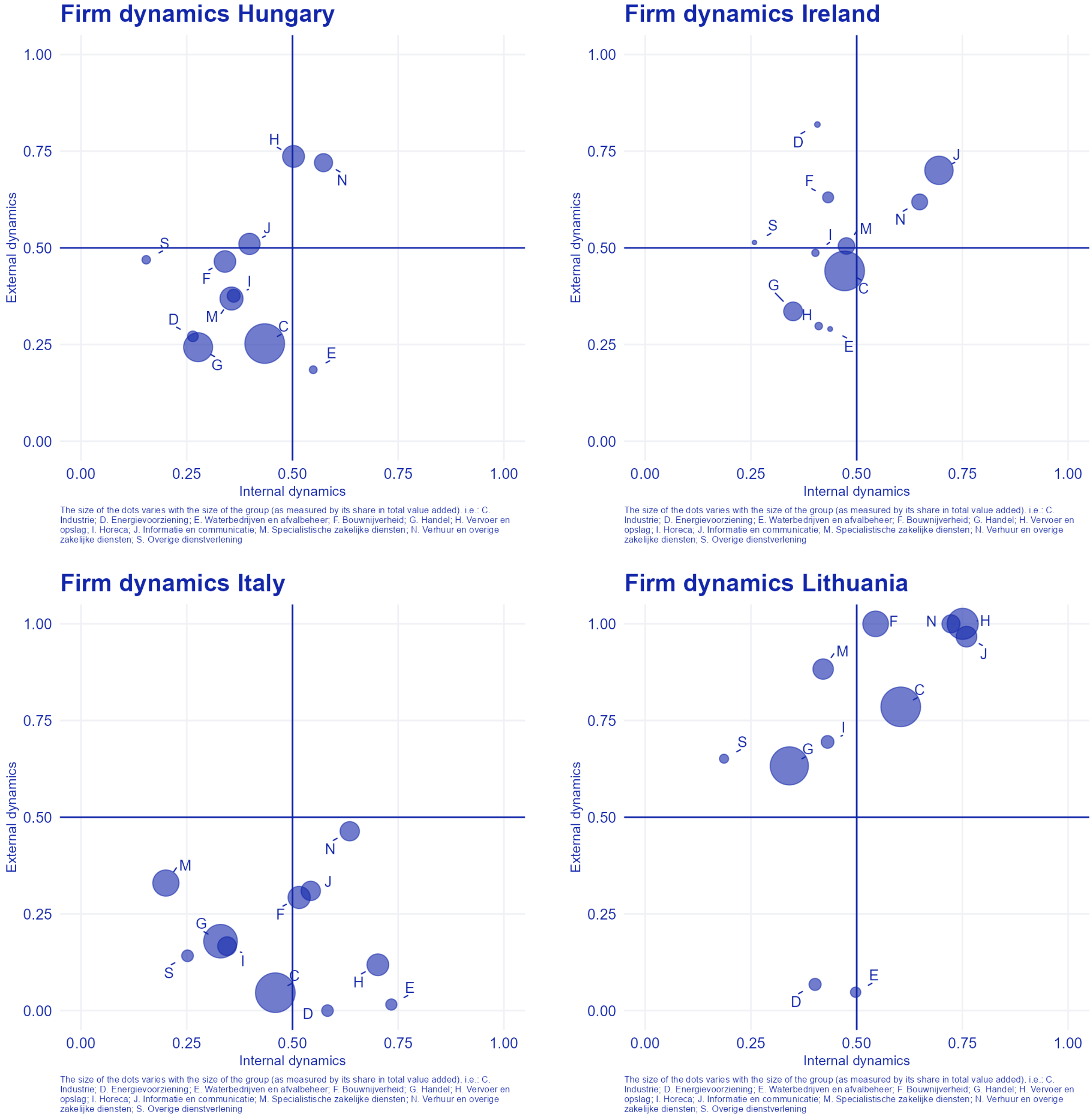

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Figure II: Internal en external dynamics dynamics per country (continued)

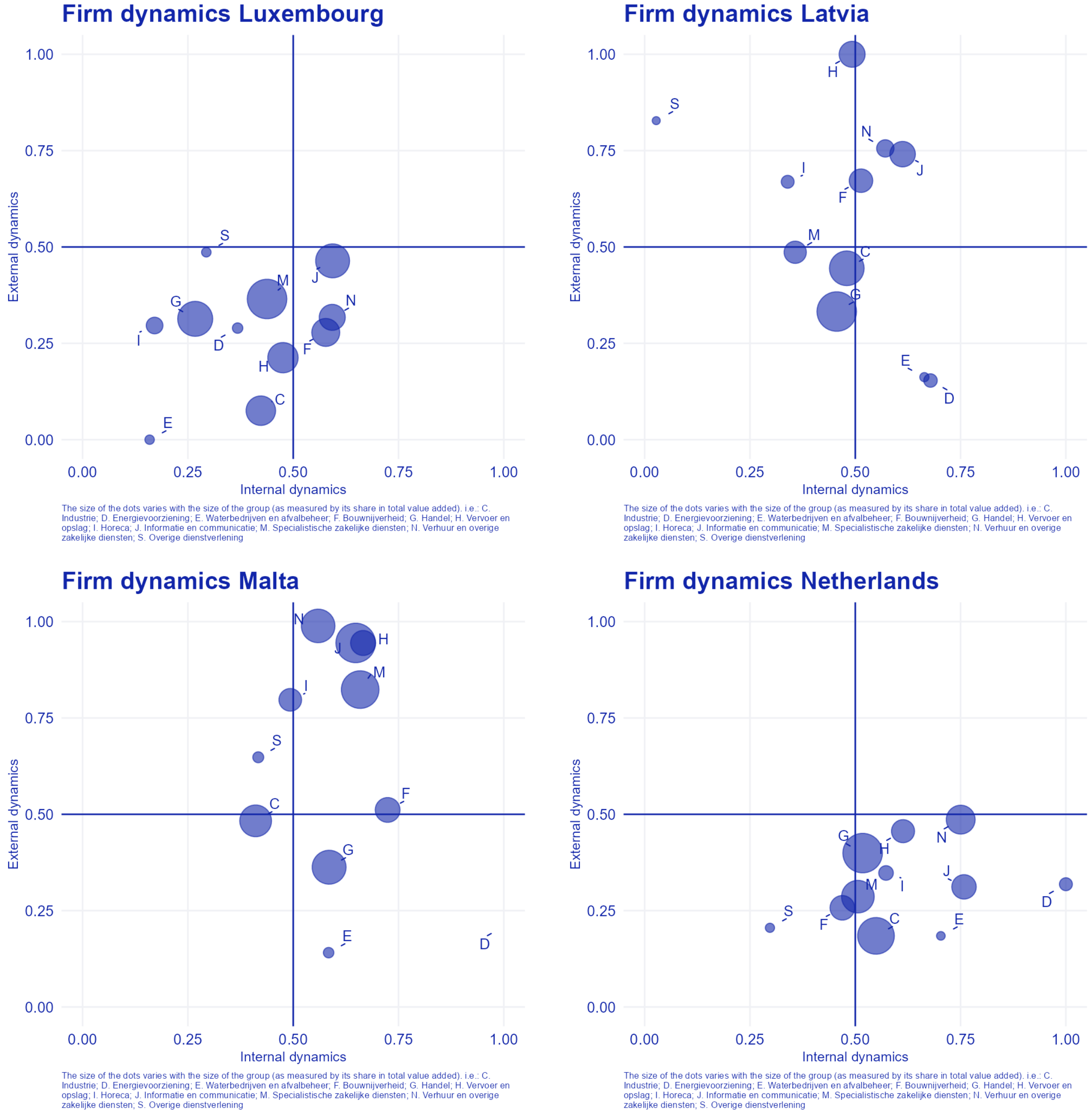

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Figure II: Internal en external dynamics dynamics per country (continued)

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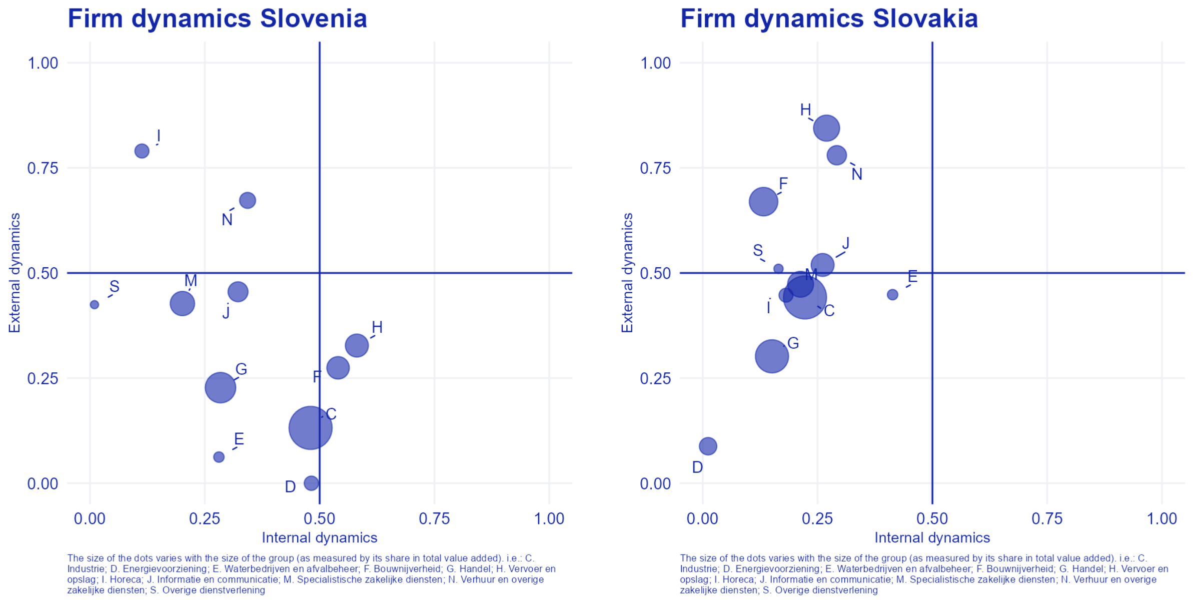
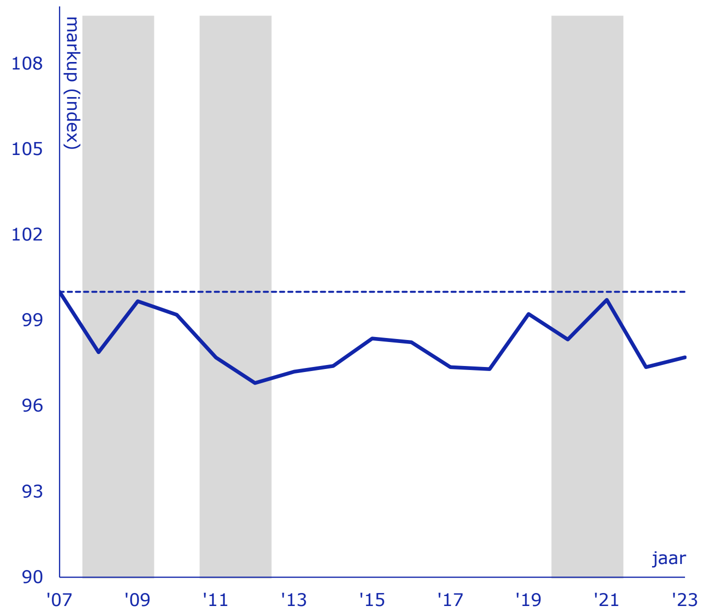
Figure II: Internal en external dynamics dynamics per country (continued)



Figuur 3: Markups in the Netherlands, method De Loecker et al. (2026)

Index (2007=100), gemiddelde gewogen, relevante marktsector.

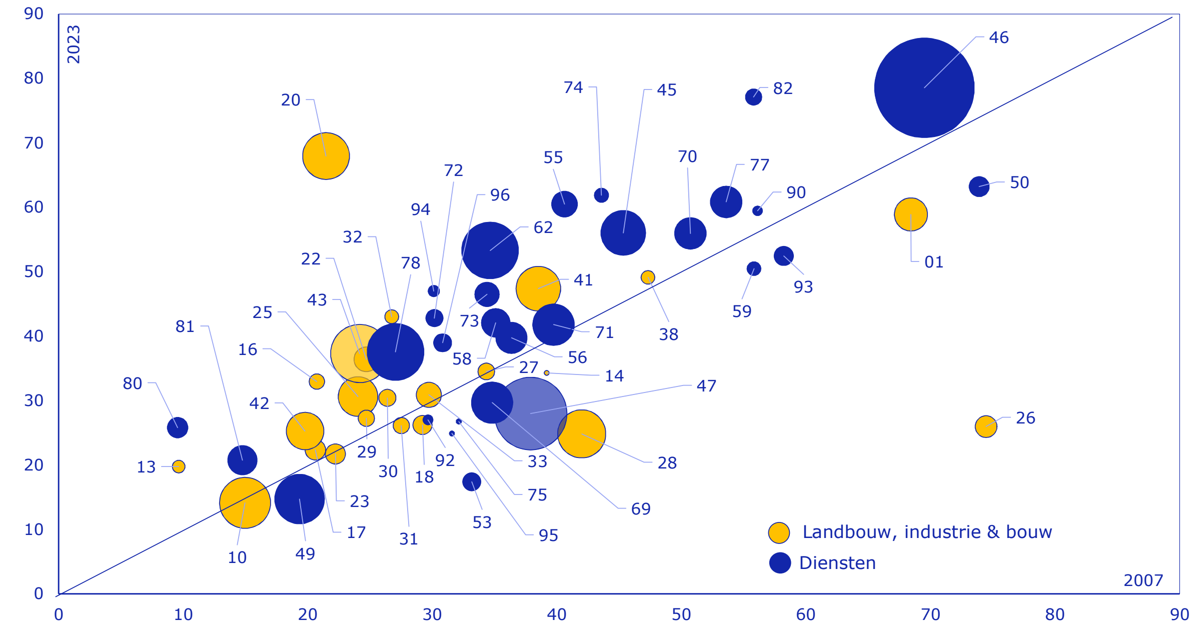


Bron: CBS-microdata, berekeningen DNB.

Toelichting: Gearceerde gebieden duiden op recessies: de financiële crisis (2008–2009), EU-schuldencrisis (2011–2012), en de COVID-pandemie (2020–2021). De blauwe horizontale stippellijn kruist de y-as bij 100. De blauwe lijn geeft de gemiddelde gewogen markup weer. De markup is berekend volgens de methode van De Loecker et al. (2026). Zie Paragraaf A.2.2 voor meer details

Figuur 4: Misallocation per industry, NACE-2D level, 2003 versus 2007

In % totale factorproductviteit (tfp), methode Hsieh en Klenow (2009)

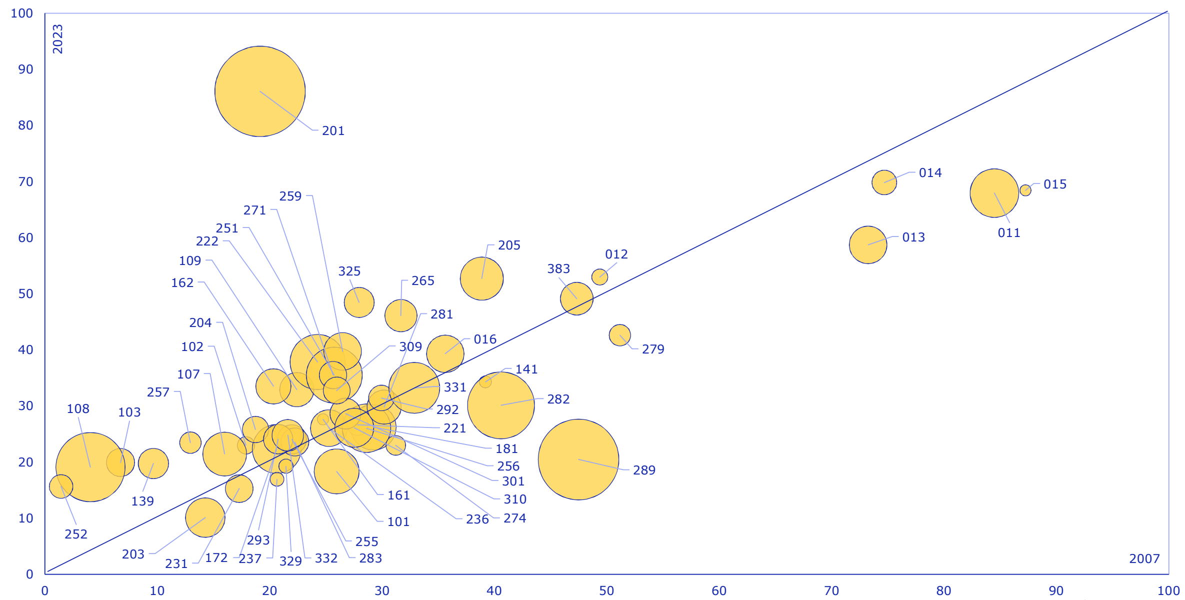


01. Landbouw
10. Voedingsmiddelenindustrie
13. Textielindustrie
14. Kledingindustrie
16. Houtindustrie
17. Papierindustrie
18. Grafische industrie
20. Chemie
21. Farmaceutische industrie
22. Kunststofindustrie
23. Bouwmaterialenindustrie
25. Metaalproducentenindstrie
26. Elektrotechnische industrie
27. Elektrische apparatenindustrie
28. Machine-industrie
29. Auto- en aanhangwagenindustrie
30. Overige transportmiddelenindustrie
31. Meubelindustrie
32. Overige industrie
33. Reparatie en installatie van machines
38. Afvalbeheer
41. Algemene bouw en projectontwikkeling
42. Grond-, water- en wegenbouw
43. Gespecialiseerde bouw
45. Autohandel en -reparatie
46. Groothandel en handelsbemiddeling
47. Detailhandel (niet in auto's)
49. Vervoer over land
50. Vervoer over water
52. Opslag, dienstverlening voor vervoer
53. Post en koeriers
55. Logiesverstrekking
56. Eet- en drinkgelegenheden
58. Uitgeverijen
59. Film, tv en radio
61. Telecommunicatie
62. IT-dienstverlening
63. Diensten op het gebied van informatie
69. Juridische diensten en administratie
70. Managementadviesbureaus
71. Architecten-, ingenieursbureaus e.d.
72. Onderzoek en ontwikkeling
73. Reclamewezen en marktonderzoek
74. Design, fotografie, vertaalbureaus
75. Veterinaire dienstverlening
77. Verhuur van roerende goederen
78. Uitzendbureaus en arbeidsbemiddeling
79. Reisbureaus, reisorganisatie en -info
80. Beveiligings- en opsporingsdiensten
81. Schoonmaakbedrijven, hoveniers e.d.
82. Overige zakelijke dienstverlening
90. Kunst
92. Loterijen en kansspelen
93. Sport en recreatie
94. Ideële, belangen-, hobbyverenigingen
95. Reparatie van consumentenartikelen
96. Overige persoonlijke dienstverlening

Misallocatie in procenten winst/verlies tfp bij efficiente allocatie. Omvang cirkels: aandeel in toegevoegde waarde relevante marksector.
Bron: CBS microdata, bewerkingen DNB.

Figuur 5: Misallocatie per industry, NACE-3D level, agriculture, industry and construction, 2003 versus 2007

In % totale factorproductviteit (tfp), methode Hsieh en Klenow (2009)

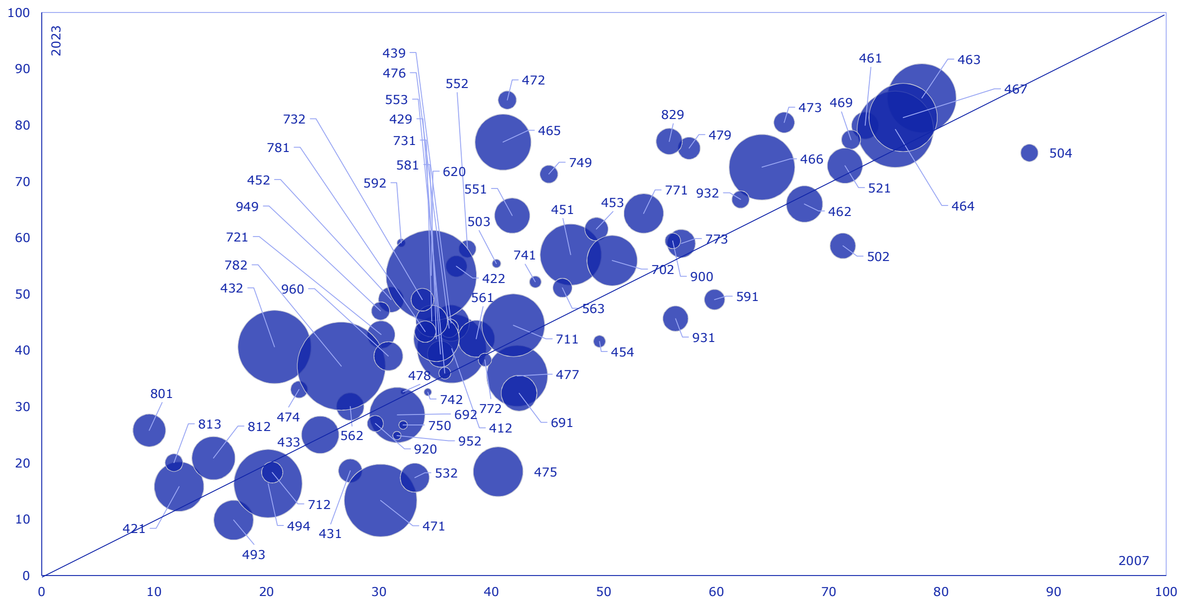


011. Teelt van eenjarige gewassen
012. Teelt van meerjarige gewassen
013. Teelt van sierplanten
014. Fokken en houden van dieren
015. Akker- en/of tuinbouw
016. Dienstverlening voor de landbouw
101. Slachterijen en vleesverwerking
102. Visverwerking
103. Verwerking van aardappels, groente en fruit
105. Vervaardigingzuivelproducten
107. Vervaardigingbrood, banketbakkerswerk
108. Vervaardigingoverige voedingsmiddelen
109. Vervaardigingdiervoeders
139. Vervaardigingoverige textielproducten
141. Vervaardigingkleding (geen bontkleding)
161. Primaire houtbewerking en verduurzaming
162. Vervaardigingartikelen van hout, kurk
172. Vervaardigingpapier- en kartonwaren
181. Drukkerijen en dienstverlening drukkerijen
201. Vervaardigingchemische basisproducten
203. Vervaardigingverf, vernis e.d.
204. Vervaardigingreinigingsmiddelen
205. Vervaardiging overige chem. producten
212. Vervaardigingfarm. Producten
221. Vervaardigingproducten van rubber
222. Vervaardigingproducten van kunststof
231. Vervaardigingglas en glaswerk
236. Vervaardigingprod. van beton/gips/cement
237. Natuursteenbewerking
251. Vervaardigingmetalen bouwproducten
252. Vervaardigingreservoirs, cv-ketels
255. Smeden, persen en profielwalsen
256. Algemene metaalbewerking
257. Industrie: eetgerei, hang/sluitwerk, gereedschap
259. Industrie: overige prod. metaal
261. Industrie: elektr componenten en printplaten
265. Industrie: meet-, regel-, navigatieapp.
271. Industrie: elektromotoren en transformatoren
274. Industrie: verlichtingsapparaten
279. Industrie: overige elektrische apparatuur
281. Industrie: motoren, turbines, en compressoren
282. Industrie: overige machines; algemeen
283. Industrie: machines en werktuigen landbouw
289. Industrie: overige machines specifiek
292. Carrosseriebouw; aanhangwagens en opleggers
293. Industrie: onderdelen en toebehoren voor auto's
301. Scheepsbouw
309. Industrie: transportmiddelen n.e.g.
310. Industrie: meubels
325. Industrie: medische instrum, en hulpmiddelen
329. Industrie: overige goederen
331. Reparatie machines en apparatuur
332. Installatie industriële machines en apparatuur
383. Voorbereiding tot recycling
411. Projectontwikkeling
412. Algemene burgerlijke en utiliteitsbouw
421. Bouw van wegen, spoorwegen en kunstwerken
422. Leggen van kabels en buizen
429. Bouw van overige civieltechnische werken
431. Slopen van bouwwerken, grondverzet
432. Bouwinstallatie
433. Afwerking van gebouwen
439. Dakbouw en overige gespec. bouw
451. Handel in auto's (icm reparatie)
452. Gespecialiseerde reparatie van auto's
453. Handel in auto-onderdelen en -accessoires
454. Handel in en reparatie van motorfietsen
461. Handelsbemiddeling
462. Groothandel in landbouwproducten /dieren
463. Groothandel in voedings- en genotmiddelen
464. Groothandel in non-food consumentenart.
465. Groothandel in ICT-apparatuur
466. Groothandel machines in industrie en handel
467. Overige gespecialiseerde groothandel
469. Niet-gespecialiseerde groothandel
471. Supermarkten en warenhuizen
472. Winkels voedings- en genotmiddelen
473. Benzinestations
474. Winkels in consumentenelektronica
475. Winkels in overige huishoudelijke artikelen
476. Winkels lectuur, sport-, kampeer- en recreatie
477. Winkels in overige artikelen
478. Markthandel
479. Detailhandel niet via winkel of markt
493. Personenvervoer over de weg
494. Goederenvervoer over de weg
502. Zee- en kustvaart (vracht-, tank- en sleepvaart)
503. Binnenvaart (passagiersvaart en veerdiensten)
504. Binnenvaart (vracht-, tank- en sleepvaart)
521. Opslag
522. Dienstverlening voor vervoer
532. Post en koeriers
551. Hotels e.d.
552. Verhuur vakantiehuisjes en appartementen
553. Kampeerterreinen
561. Restaurants, cafetaria's e.d.
562. Kantines en catering
563. Cafés
581. Uitgeverijen geschriften en databanken
591. Productie en distributie van films en tv
592. Maken en uitgeven van geluidsopnamen
619. Overige telecommunicatie
620. Dienstverlenende informatietechnologie
631. Gegevensverwerking, webhosting en -portalen
691. Rechtskundige dienstverlening
692. Accountancy, belastingadv. en administratie
702. Managementadvisering en bedrijfsvoering
711. Architecten, ingenieurs en technisch ontwerp
712. Keuring en controle
721. Natuurwetenschappelijk O&O
731. Reclamebureaus en handel in advertenties
732. Markt- en opinieonderzoekbureaus
741. Industrieel ontwerp en vormgeving
742. Fotografie en ontwikkelen van foto's en films
749. Overige special. zakelijke dienstverlening
750. Veterinaire dienstverlening
771. Verhuur en lease van auto's
772. Verhuur van consumentenartikelen
773. Verhuur/lease machines en werktuigen
781. Arbeidsbemiddeling
782. Uitzend- en uitleenbureaus en banenpools
791. Reisbemiddeling en reisorganisatie
801. Particuliere beveiliging
812. Reiniging
813. Landschapsverzorging
829. Overige zakelijke dienstverlening n.e.g.
900. Kunst
920. Loterijen en kansspelen
931. Sport
932. Overige recreatie
949. Levensbeschouwelijke en pol. organisaties
952. Reparatie van consumentenartikelen
960. Wellness, ovrige dienstverl. en uitvaart

Misallocatie in procenten winst/verlies tfp bij efficiente allocatie. Omvang cirkels: aandeel in toegevoegde waarde relevante marksector.

Figuur 6: Misallocatie per industry, NACE-3D level, services sector, 2003 versus 2007

In % totale factorproductviteit (tfp), methode Hsieh en Klenow (2009)



011. Teelt van eenjarige gewassen
012. Teelt van meerjarige gewassen
013. Teelt van sierplanten
014. Fokken en houden van dieren
015. Akker- en/of tuinbouw
016. Dienstverlening voor de landbouw
101. Slachterijen en vleesverwerking
102. Visverwerking
103. Verwerking van aardappels, groente en fruit
105. Vervaardigingzuivelproducten
107. Vervaardigingbrood, banketbakkerswerk
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204. Vervaardigingreinigingsmiddelen
205. Vervaardiging overige chem. producten
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221. Vervaardigingproducten van rubber
222. Vervaardigingproducten van kunststof
231. Vervaardigingglas en glaswerk
236. Vervaardigingprod. van beton/gips/cement
237. Natuursteenbewerking
251. Vervaardigingmetalen bouwproducten
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256. Algemene metaalbewerking
257. Industrie: eetgerei, hang/sluitwerk, gereedschap
259. Industrie: overige prod. metaal
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265. Industrie: meet-, regel-, navigatieapp.
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782. Uitzend- en uitleenbureaus en banenpools
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952. Reparatie van consumentenartikelen
960. Wellness, overige dienstverl. en uitvaart

Misallocatie in procenten winst/verlies tfp bij efficiente allocatie. Omvang cirkels: aandeel in toegevoegde waarde relevante marksector.
Bron: CBS microdata, bewerkingen DNB.

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